

## 18 Energy Balances for Tubular Reactors

### Why are we learning this?

In the previous lecture, we derived the energy balance for a CSTR. Today we will derive the energy balance for tubular reactors such as PFRs. Unlike a CSTR, conditions in a tubular reactor vary continuously with position. As fluid flows through the reactor:

- Conversion changes.
- Concentration changes.
- Temperature changes.

As a result, the reactor must be described using differential equations. By the end of today's lesson, you should understand:

- Why tubular reactors require differential energy balances.
- How heat generation and heat transfer appear in the energy balance.
- Why non-isothermal reactor design requires multiple coupled equations.
- The physical meaning of key terms in the tubular-reactor energy balance.
- The special case of adiabatic reactors.
- Why temperature and conversion must be solved simultaneously.

### Course Roadmap

Introduction to CRE



General Balance Equation



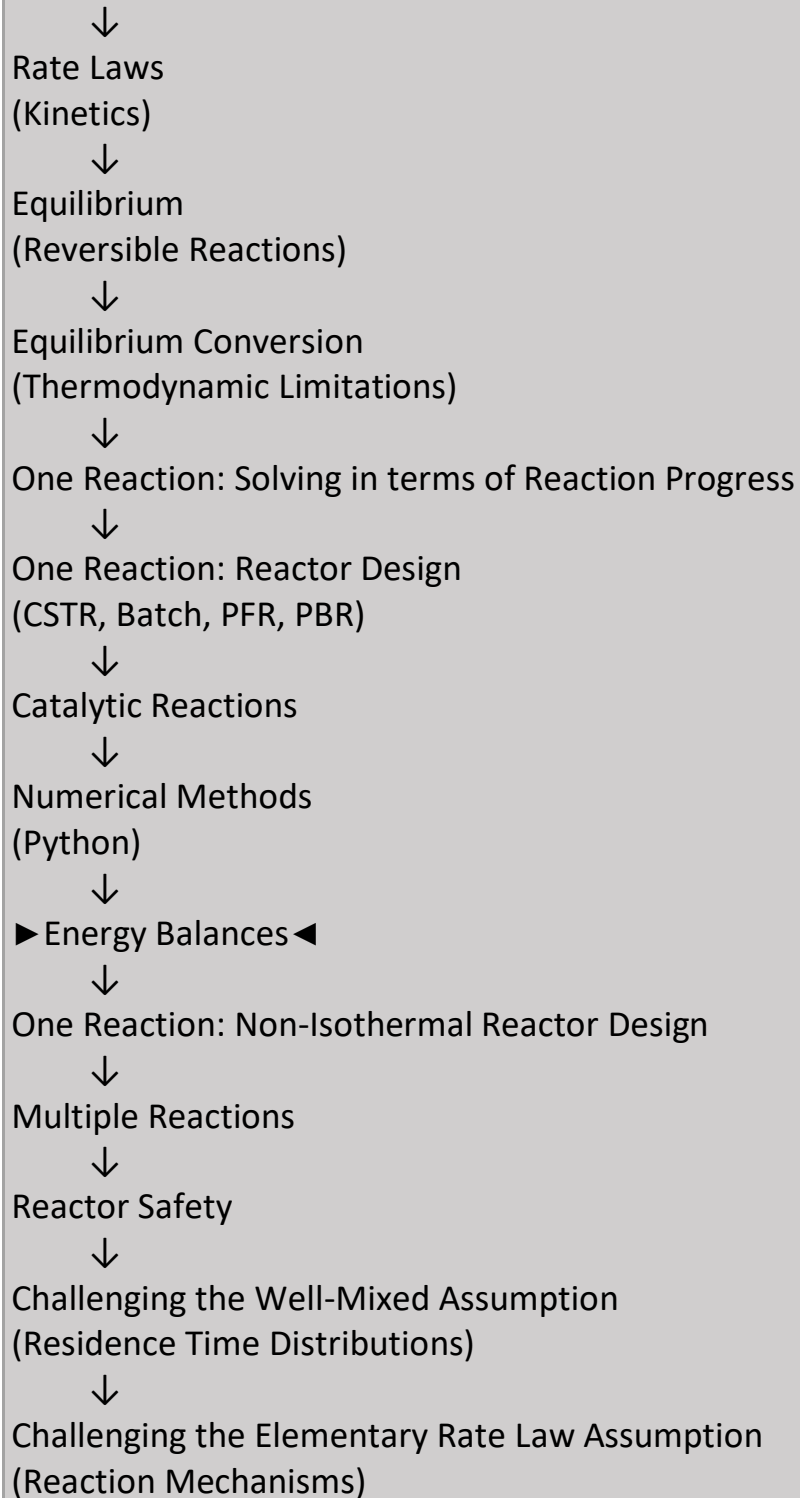
Common Reactor Models  
(CSTR, Batch, PFR)



Reaction Progress  
(Conversion)



Graphical Reactor Design  
(Inverse Rate/Levenspiel Plots)



### Big Idea

In a non-isothermal PFR:

Conversion varies with position. Temperature varies with position. Both must be solved together.

Let's derive the energy balance for a PFR:

Design equations now include:

- A mole balance in terms of  $X$  or one mole balance (in terms of  $F$ ) for each species
- An energy balance on the reactor system
- If there is pressure drop, then the design equations also include the Ergun equation
- If the temperature of the cooling fluid is not constant, then the design equations additionally include an energy balance on the cooling fluid.

### Why Are There So Many Equations?

As reactor models become more realistic, additional variables become unknown. For example:

Conversion

Temperature

Pressure

Cooling-Fluid Temperature

Each unknown requires an additional governing equation. This is why non-isothermal reactor design often involves solving several equations simultaneously.

In other words, there will be between 2 and (# of species + 1) design equations if the reactor is non-isothermal.

- If there are 2 unknowns ( $X$  and  $T$ ), you need 2 equations (mole balance and energy balance)
- If  $p$  is also unknown, you need the Ergun equation.
- If  $T_a$  (temperature of cooling fluid) varies, you need to write an energy balance on that system.
- If the problem cannot be solved in terms of  $X$ , you need to write mole balances for all species in the system.

### One Unknown Requires One Equation

A useful engineering rule: Number of Unknowns = Number of Independent Equations

Examples:

$X, T$

↓

Need Mole Balance + Energy Balance

$X, T, P$

↓

Need Mole Balance + Energy Balance + Pressure-Drop Equation

Keeping track of unknowns is often the first step in solving a reactor-design problem.

Let's derive the balance for the system pictured above:

Starting from the general form of the balance:  $0 = \dot{Q} + \dot{W}_s + \sum_i F_{i0} H_{i0} - \sum_i F_i H_i$ :

Cancel terms:

Applying the general form of the balance to the system:

$F_i$  are

$\dot{Q}$  is

Notice that the ordering inside the parentheses is intentional:

- If  $T_a > T$

- If  $T > T_a$

### Interpreting Heat Transfer

Heat transfer occurs whenever the reactor temperature differs from the surrounding temperature.

$T_a > T$

Heat enters the reactor.

$T > T_a$

Heat leaves the reactor.

The sign of the heat-transfer term therefore depends on the direction of energy flow.

UA is

In this class, we write  $\dot{Q}$  as  $Ua(T_a - T)$ , where  $a$  has units of  $\frac{\text{surface area}}{\text{volume}}$ . This is a simplification, made so that in this class we can focus on how energy balances are constructed and how they influence reactor design rather than parameterizing heat transfer coefficients.

### Why Simplify Heat Transfer?

Real heat-transfer calculations can be very complicated. In this course, we combine:

Heat Transfer Coefficient

×

Surface Area

into a single term:  $Ua$ .

This allows us to focus on the influence of heat transfer on reactor behavior rather than on detailed heat-transfer modeling.

Then:

Apply definition of the derivative to get:

Apply "product rule" to  $\sum_i F_i H_i$  term:

The first term is:

The second term is:

### **What Is Changing Inside the Reactor?**

Energy changes for two reasons:

#### **Species Amounts Change**

Reaction consumes some species and produces others.

#### **Temperature Changes**

The energy of each species changes as temperature changes.

Both effects must be included in the energy balance.

Recall that

Then:

This development assumes  $C_p$  is constant with temperature. This is a reasonable assumption to make for liquids and gases; since we deal with liquid and gas phase reaction conditions in this class, it is a reasonable assumption to make in this class.

This leaves us with the energy balance:

Since all  $r_i$  are related by stoichiometry, we can use Equation 3-1 (relative rates):

E.g., for the reaction  $A + \frac{b}{a}B \rightarrow \frac{c}{a}C + \frac{d}{a}D$ :

### **Stoichiometry Simplifies the Energy Balance**

Just as in mole balances, stoichiometry links the rates of all species. Because all species rates are related, the energy balance can ultimately be written in terms of a single reaction rate. This greatly simplifies the final equation.

Substituting these into the energy balance:

What is  $\sum_i H_i \nu_i$  equal to?

### Interpreting the Heat of Reaction

The heat of reaction represents the energy released or absorbed when reaction occurs.

$\Delta H_{rxn} < 0$  = Exothermic reaction.

$\Delta H_{rxn} > 0$  = Endothermic reaction.

This term acts as the link between reaction progress and temperature change.

We can prove this:

$\Delta H^{rxn} = \sum^{products} \nu_i H_i + \sum^{reactants} \nu_i H_i$ , where  $\nu_i$  are positive for products and negative for reactants.

For the reaction  $A + \frac{b}{a}B \rightarrow \frac{c}{a}C + \frac{d}{a}D$ ,  $\Delta H^{rxn} = \frac{d}{a}H_D + \frac{c}{a}H_C - \frac{b}{a}H_B - H_A$ .

On a per mole basis,  $\Delta H^{rxn} = \sum^{products} \frac{\nu_i}{\nu_A} H_i - \sum^{reactants} \frac{\nu_i}{\nu_A} H_i$ .

- This gives the heat of reaction in units of energy per mole of A.
- It is important to make sure that  $\Delta H^{rxn}$  is properly normalized and that it is apparent which species it is written in terms of. Just like for  $r_i$ , it is important for accounting.

Substituting in  $\Delta H^{rxn}$ :

Units check:

- $\Delta H^{rxn}$  is in units of
- $r_A$  is in units of
- The term  $(-\Delta H^{rxn})(-r_A)$  is hence in units of

Let's re-arrange so that this formula "looks like" an ODE, this will give us the final form of the energy balance:

Two **very important** terms:

- $(-\Delta H^{rxn})(-r_A)$  is
- $Ua(T_a - T)$  is

### Competing Thermal Effects

Two competing processes determine reactor temperature:

**Heat Generation:** Energy released (or absorbed) by chemical reaction.

**Heat Transfer:** Energy exchanged with the surroundings.

The balance between these effects determines the reactor temperature profile.

What if the reactor is adiabatic?

In this case, the temperature only depends on conversion. Hence:

Where  $\Delta C_P = \sum^{\text{products}} \nu_i C_{P_i} + \sum^{\text{reactants}} \nu_i C_{P_i}$ , where  $\nu_i$  are positive for products and negative for reactants.

This equation is \_\_\_\_\_ and can be used for \_\_\_\_\_

### **Adiabatic Reactors**

An adiabatic reactor does not exchange heat with its surroundings.

$$Q = 0$$

As a result: Temperature changes only because reaction occurs.

This greatly simplifies the energy balance and creates a direct relationship between conversion and temperature.

What if  $T_a$  is constant?

### **Constant Cooling-Fluid Temperature**

When the cooling-fluid temperature remains constant:

$$T_a = \text{constant}$$

the reactor energy balance is usually the only energy balance that must be solved. This situation is much simpler than the case where the cooling fluid also changes temperature.

What if  $T_a$  is not constant?

## A Second Energy Balance

If the cooling fluid heats up or cools down as it flows:  $T_a$  changes with position.

Now:

Reactor Temperature  $\leftrightarrow$  Cooling-Fluid Temperature

Both temperatures influence one another. Therefore, a separate energy balance must be written for the cooling fluid.

## Summary and Key Takeaways

Today we derived the energy balance for tubular reactors.

### Key Ideas

- ✓ Non-isothermal tubular reactors require both mole balances and energy balances.
- ✓ Conversion, temperature, pressure, and cooling-fluid temperature may all become unknowns.
- ✓ The heat-transfer term accounts for energy exchange with the surroundings.
- ✓ Stoichiometry allows the energy balance to be written in terms of the heat of reaction.
- ✓ The heat of reaction links reaction progress to temperature change.
- ✓ Adiabatic reactors are special because temperature changes arise only from reaction.
- ✓ Variable cooling-fluid temperatures require an additional energy balance.

### Most Important Idea

Reaction changes temperature. Temperature changes reaction rate. Both vary continuously throughout a tubular reactor. This coupling is the central feature of non-isothermal tubular reactor design.